FPGA Development for Radar, Radio-Astronomy and Communications

THE RADAR MASTERS COURSE

Dept. Electrical Engineering, University of Cape Town
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Convened by Dr Stephen Paine

Day 3 – 5 August 2026

Outline

Memory-Mapped Bus

Advanced Clocking

Clock Domains

Pipelines

Streaming Processors

Flow Control

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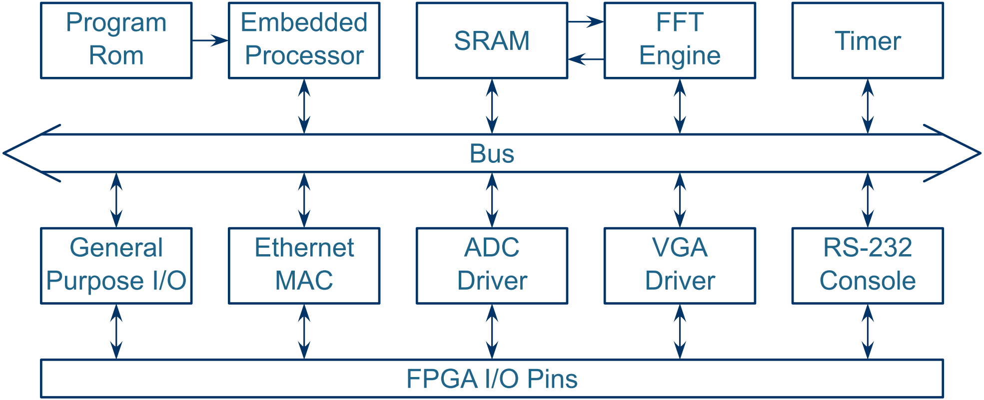
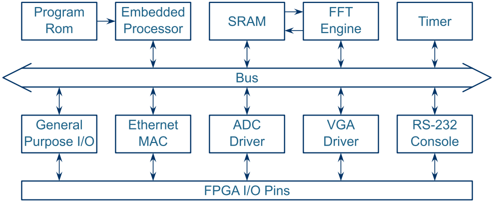
Memory-Mapped Bus

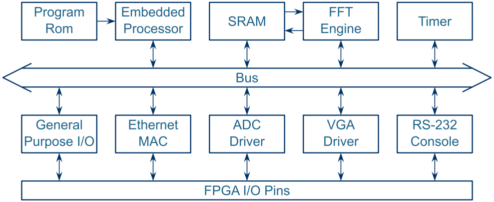


Memory-Mapped Bus



- Every node on the bus has an address range allocated
- Generally used for writing control registers, reading system status and accessing memory
- Altera Qsys uses Avalon
- Xilinx IP Integrator and ARM processors use AXI
- Many open-source projects use Wishbone

Memory-Mapped Bus



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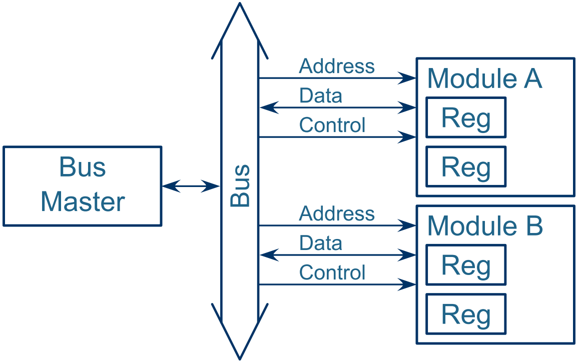
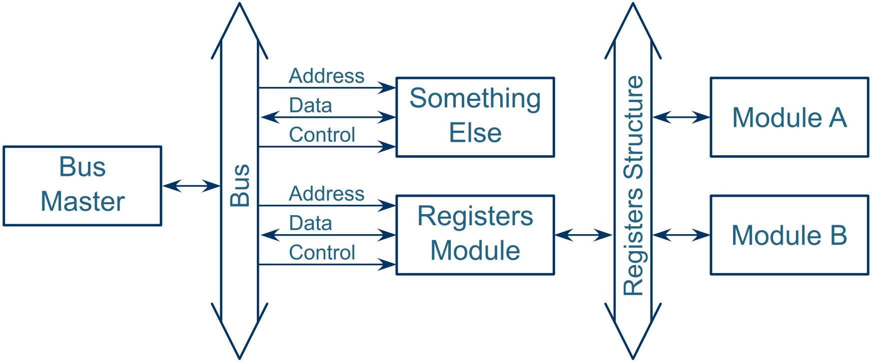
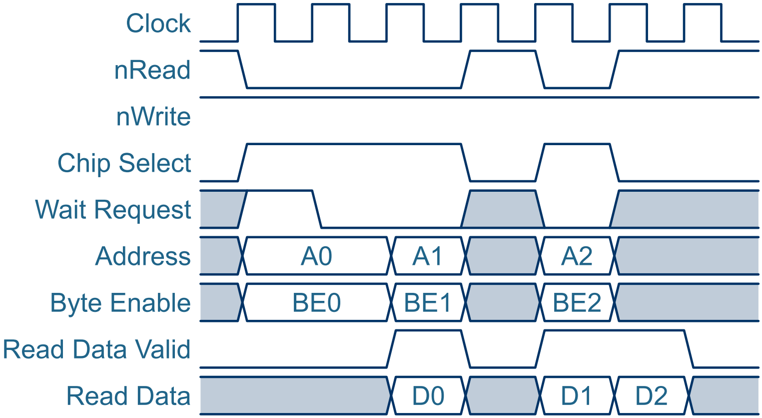
Memory-mapped Architectures



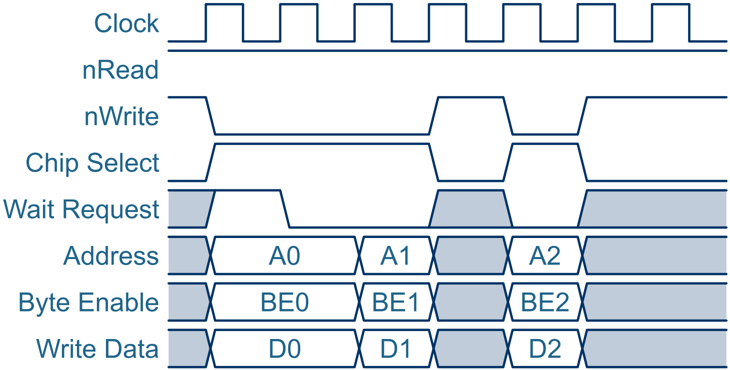
Memory-mapped Architectures



Interface Timing Diagrams



Interface Timing Diagrams



Structures

- SystemVerilog only
- Define these once in a separate file, typically as part of a package

```
package Structures;
  typedef struct{
    logic [ 1:0]Buttons;
    logic [ 9:0]Switches;
    logic [31:0]StatusBits;
  } RD_REGISTERS;

  typedef struct{
    logic [ 9:0]LEDs;
    logic [31:0]NCO_Frequency;
    logic [ 2:0]Bandwidth;
  } WR_REGISTERS;
endpackage
```

Structures

```
import Structures::*;

module Registers(
  input ipClk, ipReset,

  input  [15:0]ipAddress,
  output [31:0]opRdData,
  input  [31:0]ipWrData,
  input        ipWrEn,

  input  RD_REGISTERS ipRdRegisters,
  output WR_REGISTERS opWrRegisters
);
...
```

Structures

```
// In the top-level module...

RD_REGISTERS RdRegisters;
WR_REGISTERS WrRegisters;

Registers Registers_Inst(
  Clk, Reset,
  Address, RdData, WrData, WrEn,
  RdRegisters, WrRegisters
);

NCO NCO_Inst(
  Clk, Reset,
  WrRegisters.NCO_Frequency,
  RdRegisters.StatusBits[15]
);
```

Structures

- You can map a structure to an input port
  and then use only what you need within the submodule
- You can not map the same structure to an output port of more than one module,
  i.e. you need to map the structure members individually
- Luckily, most registers are control registers,
  and therefore input ports of the target modules

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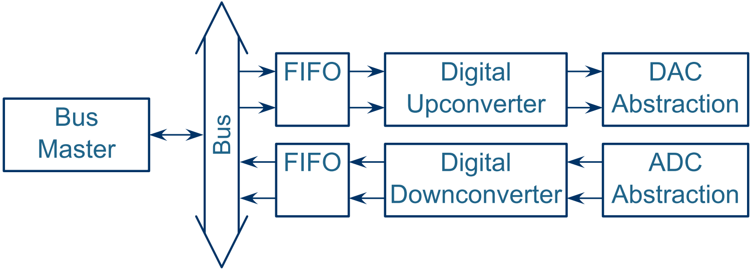
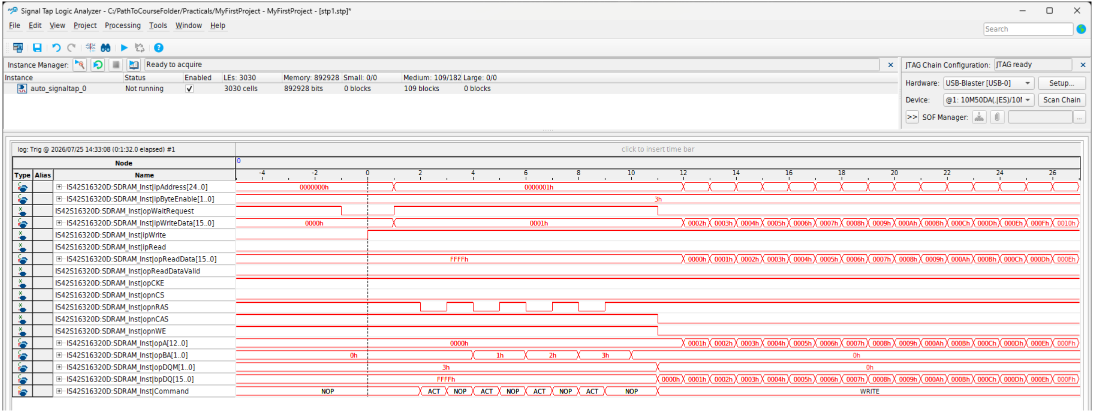
Memory-mapped Streams



- It is often convenient to allocate a bus address to a stream
- A write to that address injects one unit into the stream
- A read from that address reads one unit from the stream
- Most often unidirectional and FIFO-buffered
  with feed-back registers

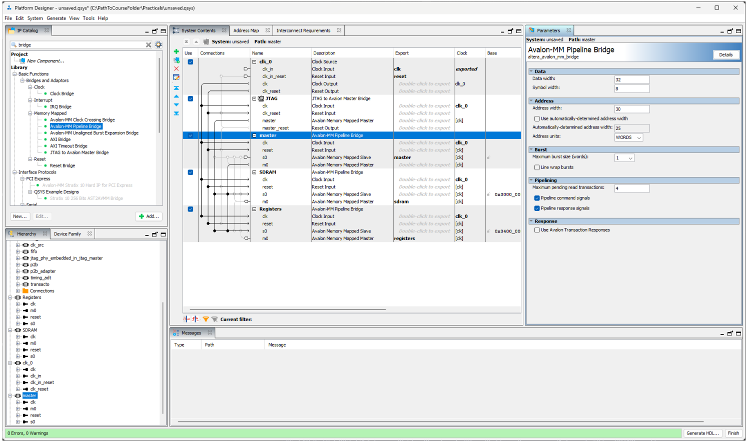
Platform Designer

Practical `05 – SDRAM` hooks up the SDRAM and introduces Signal Tap



Platform Designer

Practical `06 – Platform Designer` introduces the Platform Designer



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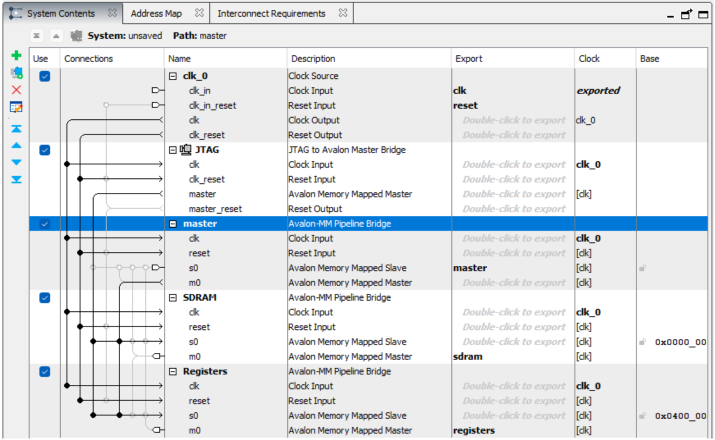

Outline

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Advanced Clocking

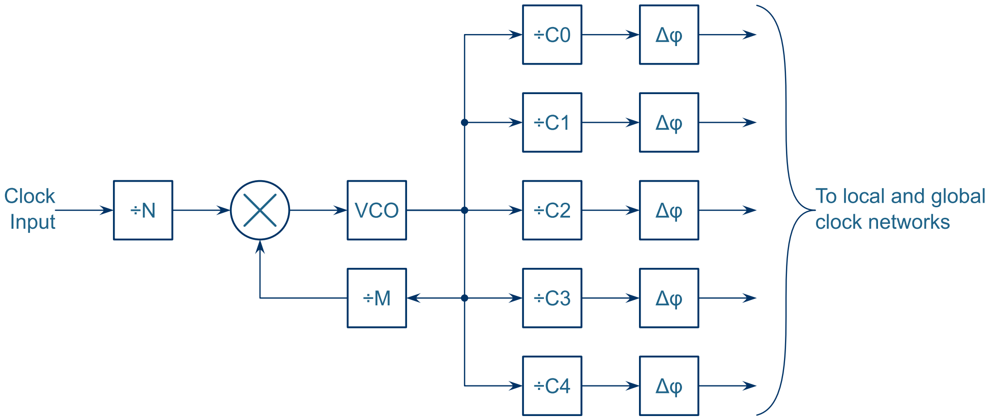
Clock Domains

Pipelines

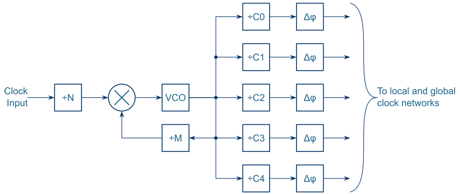
Streaming Processors

Flow Control

Phase-locked Loop

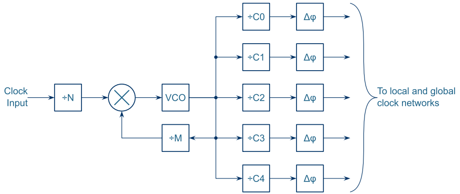


Phase-locked Loop



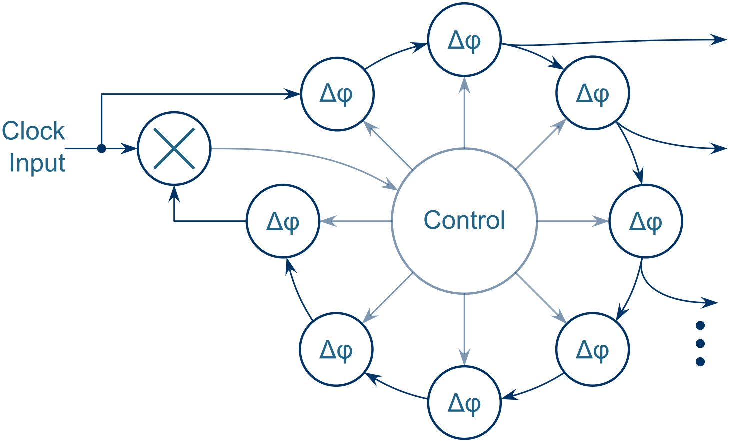
- Useful for generating a range of related clocks
- Output clock rising and falling edges can be set, independently, to a resolution equal to the VCO period, which is typically about 1 ns $\pm$ 500 ps
- All output clocks can be considered to be in the same clock domain
- Noise sensitive applications should avoid PLLs due to high phase noise

PLL Compensation



- **Direct**: No compensation, which results in better jitter performance
- **Normal**: The clock at the register is phase-aligned with the input clock pin
- **Source Synchronous**: The PLL ensures that the data delay from pin to register and the apparent clock delay from pin to register is the same
- **Zero-Delay Buffer**: The output clock pin is phase-aligned with the input clock pin

Delay-locked Loop



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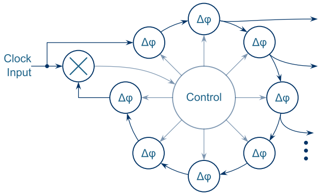
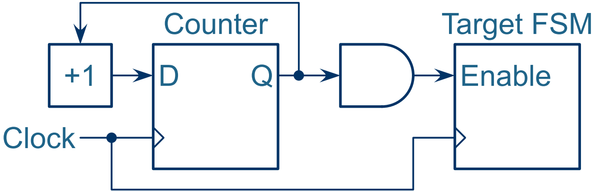
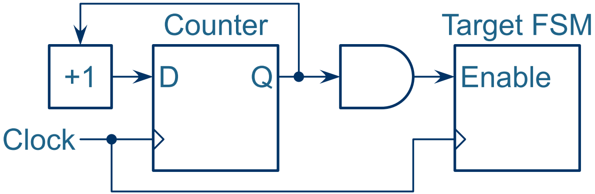


- Useful for performing phase compensation when interfacing to fast peripherals
- All output clocks can be considered to be in the same clock domain as the input clock

Clock-Enable Generated

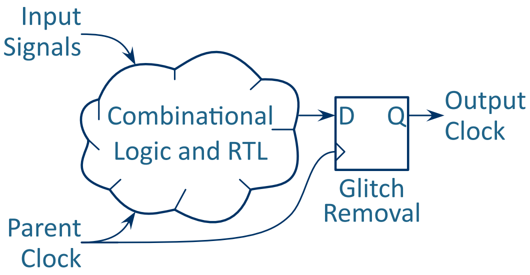


Clock-Enable Generated

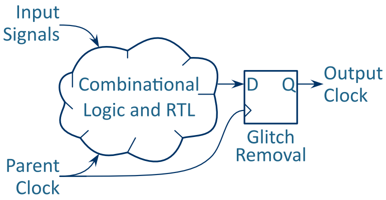


- Useful when mixing fast and slow systems
- Potential to reduce overall device power usage
- Cannot be used for clocking external devices
- The multi-cycle timing requirements must be manually specified

Ripple and Gated Clocks

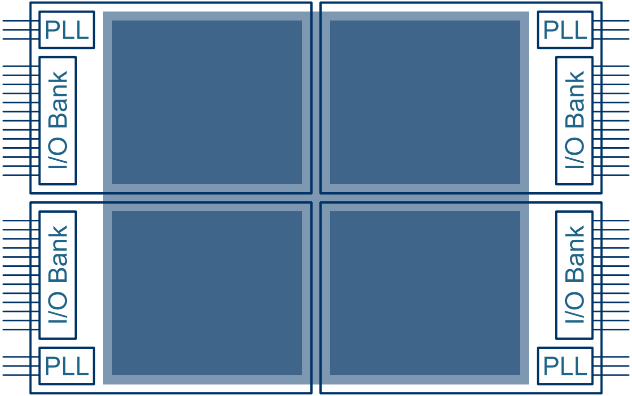


Ripple and Gated Clocks

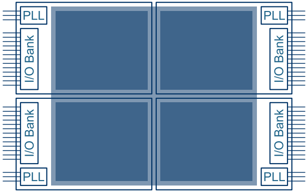


- Unrelated to all other clocks in the system, including the parent clock
- Often necessitates complex clock-domain crossing schemes to the general system clock
- Useful when:
  - Requiring very low power usage
  - The frequency must change often
  - The clock must drive an external device
  - etc.

Clock Regions



Clock Regions



- FPGAs are generally organised into regions, each with its own PLL and I/O bank
- The local clock network is more efficient (and faster) than the global network
- When laying out the PCB, keep fast I/O in the same region as their clock

Outline

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Flow Control

Common Mistake

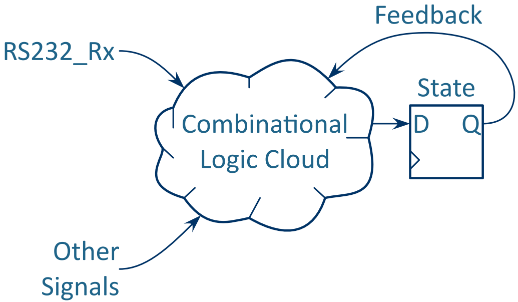
```
module RS232(
  input  ipClk, ipReset,
  input  ipRS232_Rx, // Asynchronous external signal
  output [7:0]opData
);

... // Definitions, state machine boilerplate, etc.

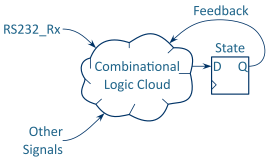
  Idle: begin
    if(!ipRS232_Rx) begin
      State <= Receiving;
    end
  end

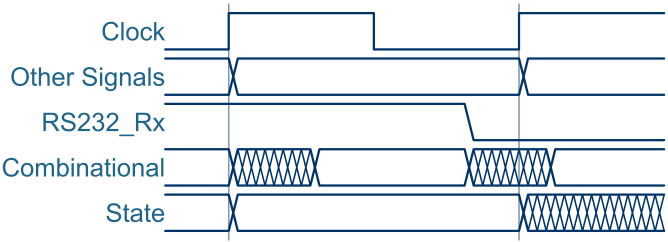
  Receiving: begin
...
```

Common Mistake

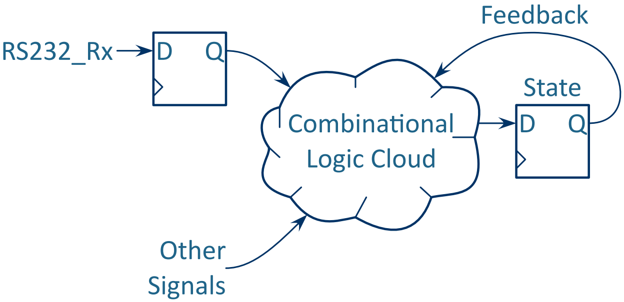


Common Mistake





The Solution



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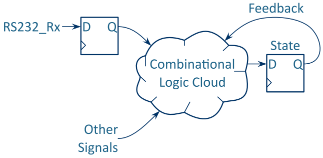
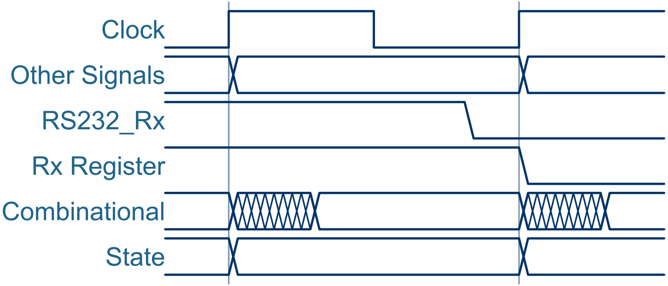
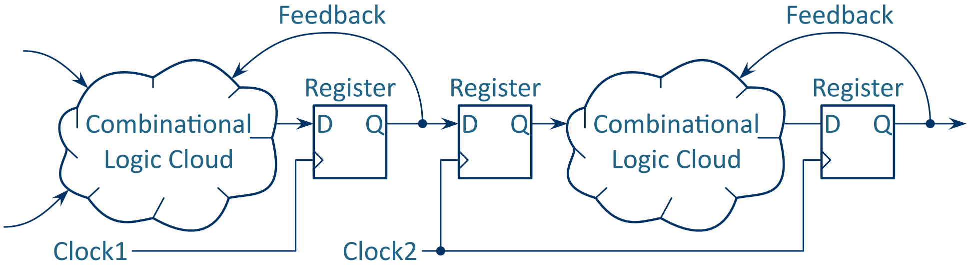
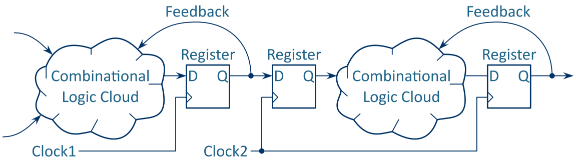




Register Chain

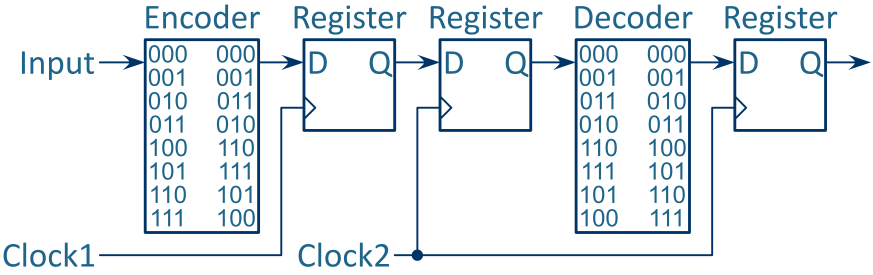


Register Chain

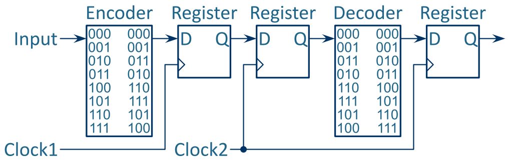


- Essential for crossing asynchronous external signals
- Only useful for crossing one bit at a time (unless the system is tolerant to temporary errors)
- Often used to cross hand-shaking signals
- Works in both directions (fast to slow / slow to fast)

Gray Coding

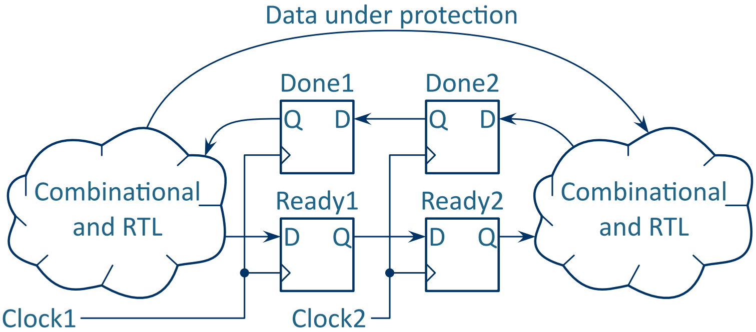

Encoder
Register
Register
Decoder
Register
Input
000 000
001 001
010 011
011 010
100 110
101 111
110 101
111 100
D Q
D Q
000 000
001 001
011 010
010 011
110 100
111 101
101 110
100 111
D Q
Clock1
Clock2

Gray Coding



- Encoded such that only one bit changes at a time
- Only works for counter data
  (FIFO queue pointers, rotary encoders, etc.)
- Works in both directions (fast to slow / slow to fast)

Hand-Shaking



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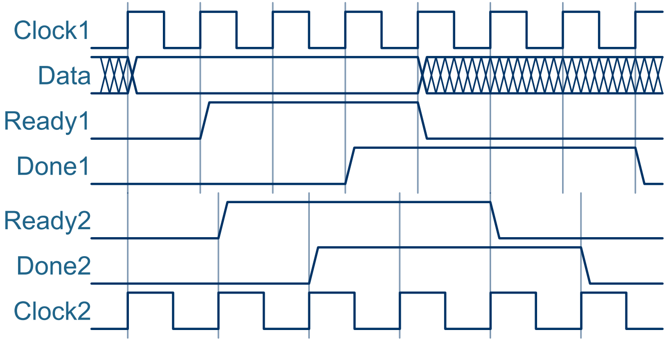

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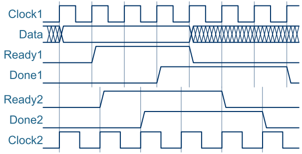

- Source must guarantee data stability while the handshaking is taking place
- Not time-efficient: need to wait for the entire transaction sequence to finish before starting the next one
- Use only for data that does not need to be crossed often
- Works in both directions (fast to slow / slow to fast)

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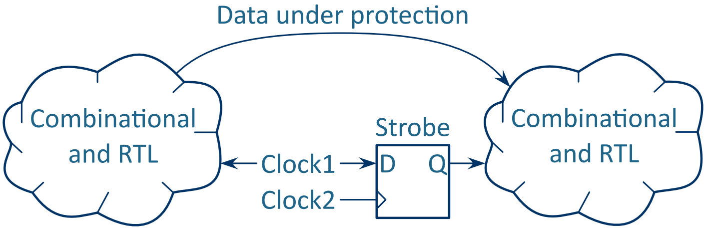
Strobe-based



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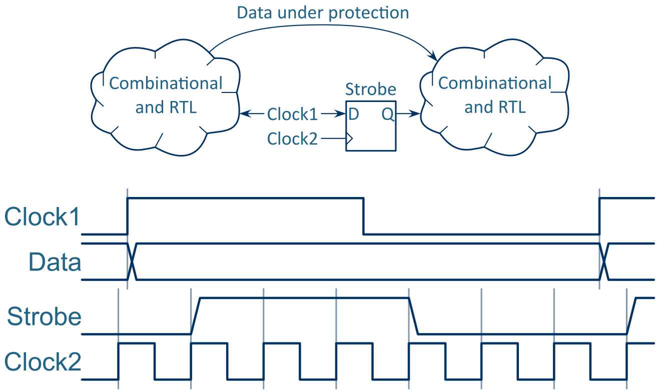
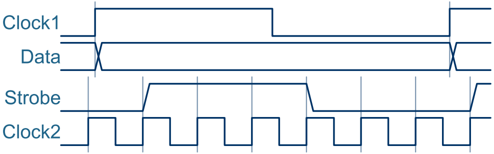


Strobe-based



- The data is valid for as long as the strobe is high
- As fast as the source clock
- The strobe edges can be used to perform flow-control
- Only works when the destination clock is much faster than the source clock
- Note that the source “clock” can be a strobe signal generated by the source state machine...

FIFO Queues

- Flow-control signals are crossed separately
- Control signal crossing latency can be hidden in the length of the queue
- Mixed-width RAM can be used to keep the data-rate constant across different frequencies (especially useful in DDR-based external memory)

Clock Groups

- Unrelated clock domains must be kept separate in the timing constraints as well

```
set_clock_groups -logically_exclusive \
  -group [get_clocks ADC_CLK_10]       \
  -group [get_clocks MAX10_CLK1_50]    \
  -group [get_clocks MAX10_CLK2_50]    \
  -group [get_clocks {opClk_SDRAM *altpll_component*}]
```

Coffee Break...

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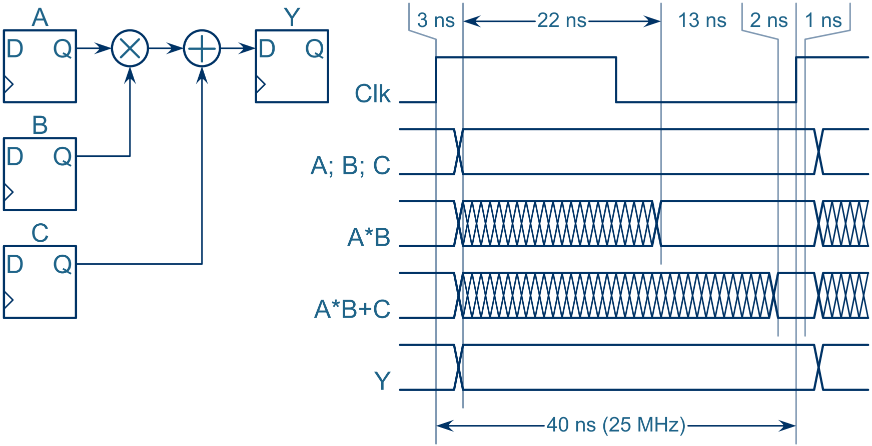
Simple Pipelines



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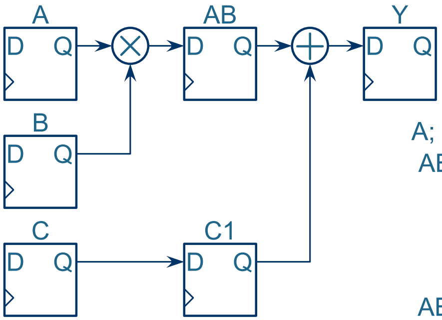

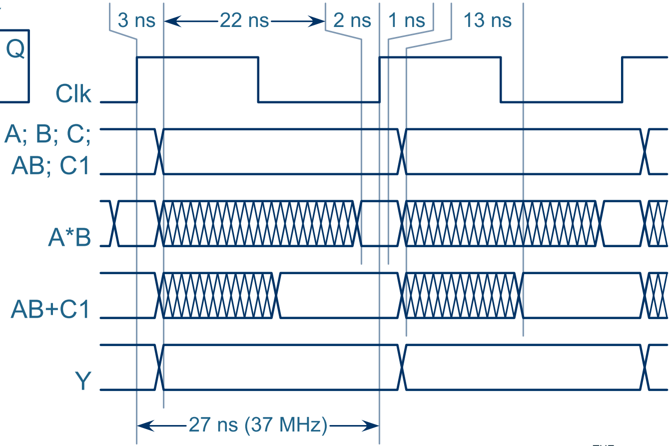

Simple Pipelines

- Gain throughput at the cost of latency and resources
- Often easier to use arrays, especially with long chains
- Keep the index equal to the stage which assigns the value

```
always @(posedge ipClk) begin
  // Stage 1
  AB <= A*B;
  C1 <= C;

  // Stage 2
  Y <= AB + C1;
end
```

Simple Pipelines

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```
reg [7:0]C[1:0];

always @(posedge ipClk) begin
  // Stage 0
  C[0] <= C_Input;

  // Stage 1
  AB   <= A*B;
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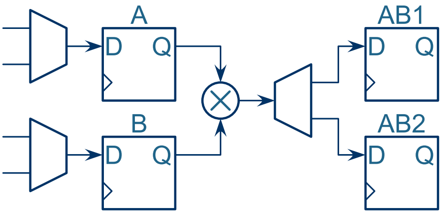
end
```

Sharing Resources

```
reg  [ 7:0]A, B;
wire [15:0]AB = A * B;

State1: begin
  AB2   <= AB;
  A     <= A1;
  B     <= B1;
  State <= State2;
end

State2: begin
  AB1   <= AB;
  A     <= A2;
  B     <= B2;
  State <= State1;
end
```



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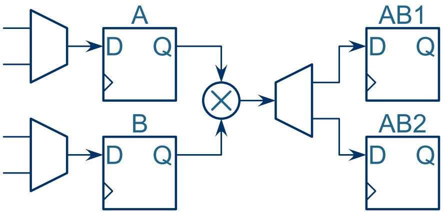
```



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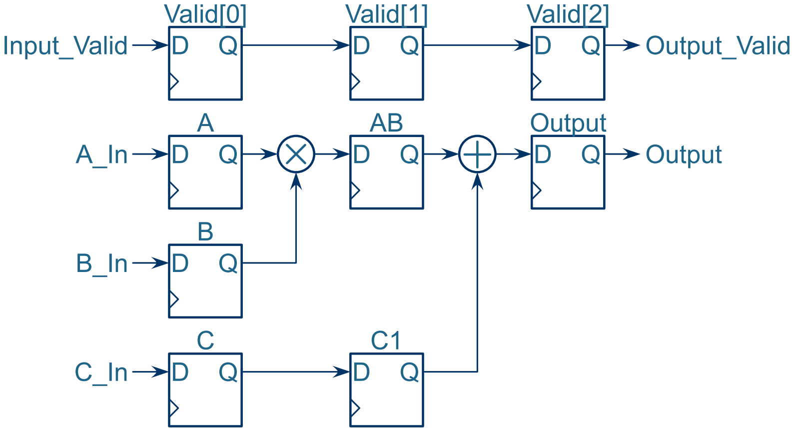
Stream Pipeline



Stream Pipeline

```
reg [2:0]Valid;

always @(posedge ipClk) begin
  // Input
  A      <= A_In; B <= B_In; C <= C_In;
  Valid <= {Valid[1:0], Input_Valid};

  // Stage 1
  AB <= A*B;
  C1 <= C;

  // Stage 2
  Output <= AB + C1;
end

assign Output_Valid = Valid[2];
```

Back-pressure

```
reg [2:0]Valid;

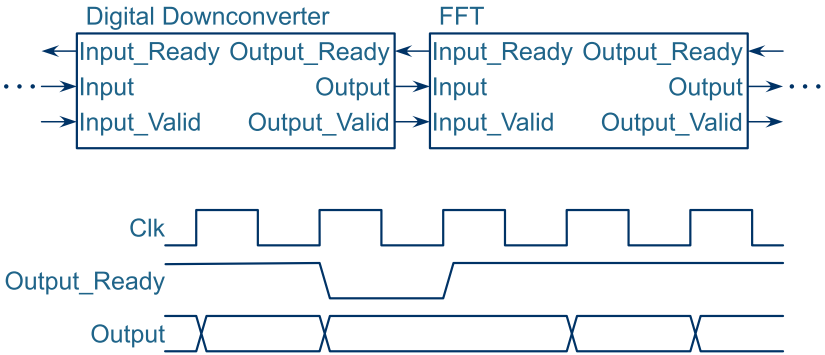
always @(posedge ipClk) begin
  if(Output_Ready) begin
    // Input
    ...

    // Stage 1
    ...

    // Stage 2
    Output <= AB + C1;
  end
end

assign Input_Ready  = Output_Ready;
assign Output_Valid = Valid[2];
```

Streaming Processor



Handshaking Strategies

Different buses have different handshaking strategies:

- An Avalon source is allowed to wait for the sink to be ready before asserting the `Valid`.
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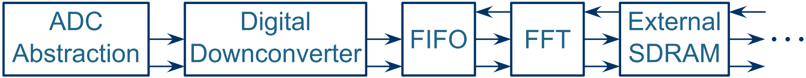
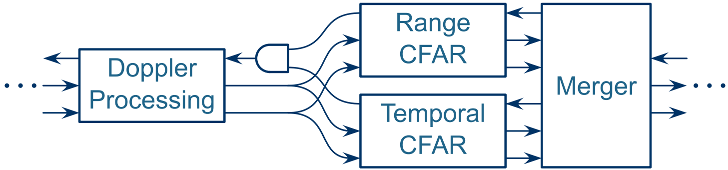
FIFO Queues



- Back-pressure makes it possible to move all the FIFO queues into a single queue
- Generally put the queue where the least amount of data is (saves resources)
- DSP chains have processing gain, so place the FIFO early in the chain

Complex Structures



- Gate the “Valid” with the “Ready”
  (if either sink is not ready, both sinks must see a “not valid” input)
- The “Merger” can take various forms:
  - Interleave
  - Alternate (sent one logical unit from the one, then the other, then repeat)
  - Combine through calculation
  - etc.

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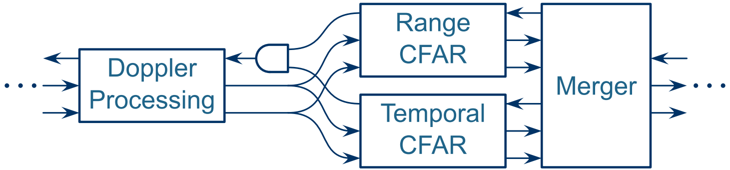

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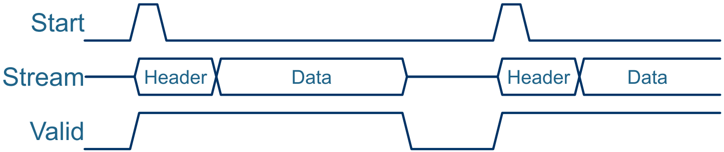
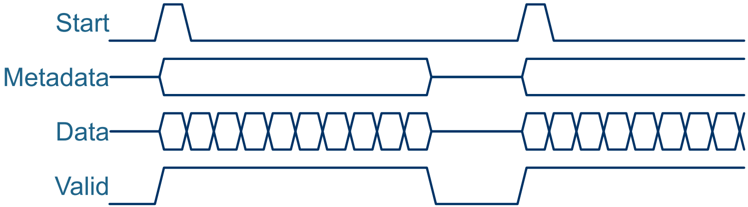
Packet-based Streams



- Add a start-of-packet strobe (valid for one clock-cycle only)
- Natural fit for RADAR: one packet per PRI
- Natural fit for packet-based communication, e.g. Ethernet or UDP (makes it easy to implement Ethernet-based FPGA-in-the-loop testing)
- The header can contain all sorts of metadata...

Packet-based Streams



- Metadata can also be implemented as a side-channel
- The metadata is stable for the duration of the packet
- No need to extract and / or add the header
- Easier to filter the stream

Outline

Memory-Mapped Bus

Advanced Clocking

Clock Domains

Pipelines

Streaming Processors

Flow Control

Fire and Forget

- No feedback from the destination to the source
- Assumes that the destination will handle the data, or
- The application can accept data loss

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- The source aims to fill the window with data
- This is used in TCP, among other protocols

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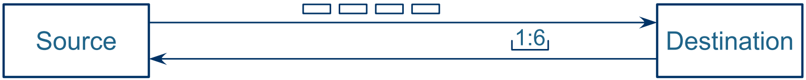

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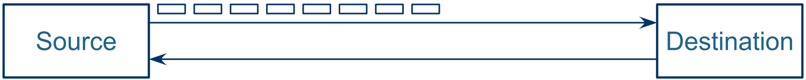

- The source assumes that the destination has a certain space
- The destination acknowledges all incoming packets
- The source sends packets aiming to keep the credit on zero
- Works best when the credit is large enough to keep both directions full at all times

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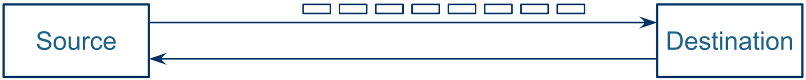

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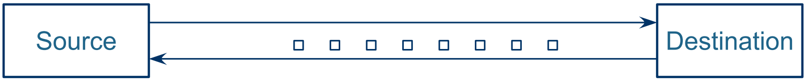

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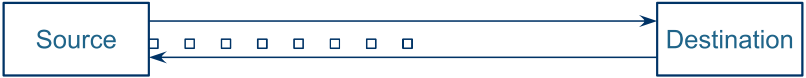

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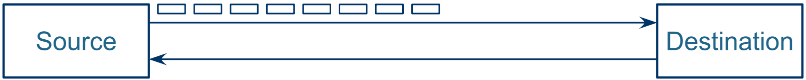

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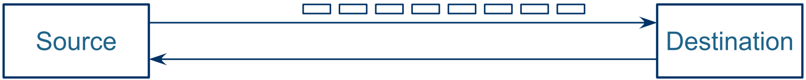

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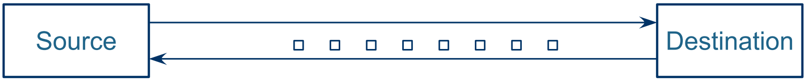

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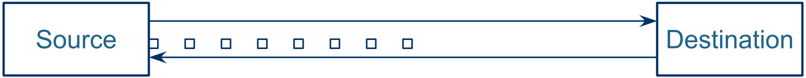

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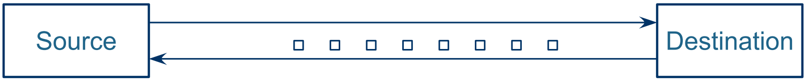

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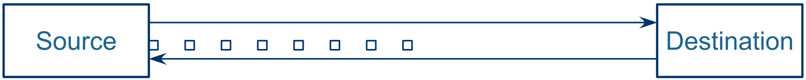

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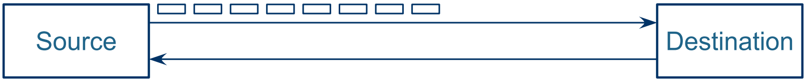

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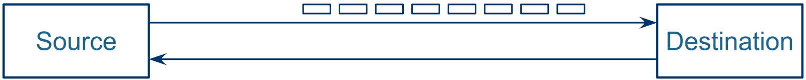

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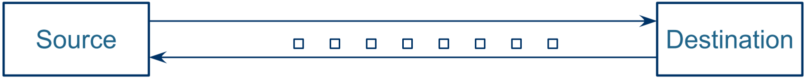

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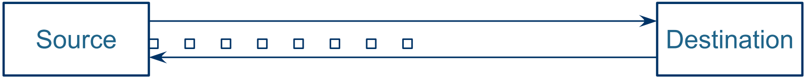

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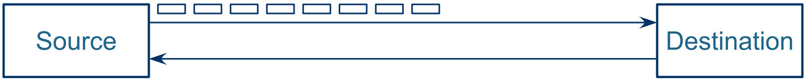

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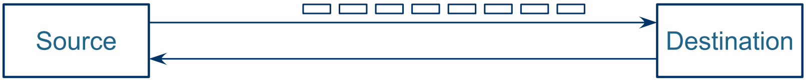

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FPGA Development for Radar, Radio-Astronomy and Communications

THE RADAR MASTERS COURSE

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Private Bag, Rondebosch, 7701, South Africa
http://www.rrsg.uct.ac.za

Presented by Dr John-Philip Taylor

Convened by Dr Stephen Paine

Day 3 – 5 August 2026